Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.–3)

PHARMACEUTICAL ORGANIC CHEMISTRY-II

Subject Code : BP-301T

M.Code :75105

Date of Examination : 02-12-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.

SECTION-A

1. Answer briefly:

a) Who proposed resonance structure of benzene? What is its bond length?

b) Name catalyst used in the nitration of benzene and Friedel craft alkylation.

c) What is saponification?

d) Give one derivative of phenanthrene used in medicine.

e) Why is aniline less basic than ethylamine?

f) Give the structure of o-cresol and one use.

g) Why is benzoic acid soluble in NaOH?

h) What is the difference between Hydrogenation and hydrolysis of oils?

i) What is the ideal bond angle for strain-free rings?

j) What happens when cyclobutane is treated with bromine?

Chandigarh College of Pharmacy LIBRARY Landran, Mohali

(S29)-848

NOV 202

SECTION-B

2. Discuss in detail the stability of cycloalkanes with reference to Baeyer's strain theory, Sachse-Mohr's theory and angle and torsional strain.

3. Explain the mechanism of nitration, sulphonation and halogenation of benzene.

4. Discuss the synthesis, chemical reactions and medicinal uses of any two polynuclear hydrocarbons.

SECTION-C

5. Define basicity. Compare the basicity of aliphatic and aromatic amines.

6. Explain the significance of acid value, iodine value, and saponification value.

7. Describe the structure and medicinal/chemical uses of DDT, BHC and Chloramine.

8. Discuss structure and reactivity of diphenylmethane and triphenylmethane.

9. Discuss the acidity of phenols in detail. How do different substituents affect it?

10. Describe the synthetic uses of aryl diazonium salts.

11. Write the mechanism of Friedel-Crafts alkylation of benzene. Mention its limitations.

12. Describe the reactivity of cyclopropane and cyclobutane with examples.

13. Explain the synthesis and reactions of naphthalene.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.